EXNO: 3

Designing a Quarter wave transformer by varying following parameters

a) frequency

b) input impedance

c) characteristic impedance

Test Case 1:

Design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

OUTPUT:

ZR=100 ohms Zin =50 ohms

ZS ohms	ZR ohms	Ro ohms
100	50	70.71

ZR=100 ohms Zin =50 ohms

Frequency (MHz)	Length of λ/4 transformer (meters)
100	0.750
200	0.375
300	0.250
500	0.150

Simulation Results:

Scilab code:

```
Zr=100;s=9;d=0.1;
Zs=50;
Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;
Zof=276*log10(s/r);
Zoq=sqrt(Zr*Zof);
L=300/(f*(10^-6));
printf("length of lambda/4 transformer = %f meters\n",L);
printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```

Result:

```
length of lambda/4 transformer =
0.600000 meters
characteristic impedance of quarter
wave line = 70.710600
```

Execute Reset

Report bug / Give Feedback

Theoretical Results

$f = 100MHz$

$$L = \frac{3\times10^8}{4(100\times10^6)} = 0.75m$$

$f = 200MHz$

$$L = \frac{3\times10^8}{4(200\times10^6)} = 0.375m$$

$f = 300MHz$

$$L = \frac{3\times10^8}{4(300\times10^6)} = 0.25m$$

$f = 500MHz$

$$L = \frac{3\times10^8}{4(500\times10^6)} = 0.15m.$$

Test Case 2:

For a fixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Zin) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance (Ztransformer= Ro= sqrt(Zin·ZL) changes with Zin.

OUTPUT:

ZR=100 ohms Variable: Zin

Zin	**Ro**
25	**50**
50	**70.71**
75	**86.60**
100	**100.00**

Simulation Results: